

Proving Two Arithmetic Conjectures for OEIS A321941

A nonlinear product identity, integral polynomial deformations,
sharp denominator bounds, and asymptotic inversion

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Abstract

Brent, Glasser, and Guttmann introduced rational coefficients d_k in the all-orders asymptotic expansion of a Hadamard product involving the exponential integral. They conjectured that $r_k = 64^k d_k$ is always an integer, and observed that $r_k \equiv \binom{2k}{k} \pmod{32}$ for $k \geq 3$. This article proves both arithmetic assertions. The key is to retain a quadratic invariant of the underlying second-order recurrence instead of using only its linear symmetric-square recurrence. After scaling, the invariant gives a triangular nonlinear equation with integer shift operators. An evenness induction removes all coefficient-index divisions.

The construction extends to polynomials $P_k(t) \in \mathbb{Z}[t]$ with $P_k(1) = r_k$, $P_k(t) \in 2\mathbb{Z}[t]$ for $k > 0$, and a coefficientwise congruence modulo 32. Both the leading coefficient and the constant term of P_k have closed forms. Consequences include the minimality of the scale 64, sharp denominator bounds at infinitely many indices, and exact denominator formulas whenever the index has at most four nonzero binary digits. Finally, the product expansion is inverted to all algebraic orders; after normalization, the inverse coefficients have no denominator primes other than 3.

The analytic existence of the original expansion is the published theorem of Brent–Glasser–Guttmann. The new arithmetic arguments are proved here. Their separate negativity conjecture is not proved. Exact finite checks accompany the article but do not replace its proofs.

Review status. This is an original research draft with human-readable proofs and executable checks, not a peer-reviewed publication or a Lean/Rocq formalization. The literature and repository comparison is targeted, not exhaustive. No repository files or OEIS entries have been modified.

[Write note, batch 7301: filed in ProveIt’s research-report collection; provenance and status in Section 1.2.]

[Write note, batch 86A, 4 October 2026: the report now has two Parts. Part II (Sections 13–21, Appendices C–D), from a later manuscript, proves the negativity conjecture $r_k < 0$ for every $k \geq 3$ and the large-order law of r_k ; see Section 13.1.]

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Part I

Integrality, congruences, and inversion

1 The problem and the results

Define

$$f_0(z) = \exp\left(\frac{z}{1-z}\right), \quad f_1(z) = \exp\left(\frac{1}{1-z}\right) E_1\left(\frac{1}{1-z}\right), \quad (1.1)$$

where $E_1(x) = \int_x^\infty e^{-u} u^{-1} du$ for $x > 0$, and write

$$a_n = [z^n]f_0(z), \quad b_n = [z^n]f_1(z), \quad \rho_n = a_n b_n.$$

These sequences arise naturally from special functions and combinatorics. For example, $n!a_n$ counts partitions of an n -element labelled set into nonempty linearly ordered blocks, the classical sequence A000262 [3]. The second sequence is a recessive companion of the same second-order recurrence. Their product cancels the opposite stretched exponentials and exposes a more subtle arithmetic structure.

Brent, Glasser, and Guttmann proved that

$$\rho_n \sim -\frac{1}{2n^{3/2}} \sum_{k \geq 0} d_k n^{-k}, \quad d_0 = 1, \quad d_k \in \mathbb{Q}. \quad (1.2)$$

Here and throughout, an all-orders expansion is meant in the Poincaré sense: for each fixed $M \geq 0$, truncation after d_M has normalized remainder $O(n^{-M-1})$. We use the analytic existence statement from their Corollary 9; we do not claim to rediscover it [1].

Their Conjecture 10 asserts that $r_k = 64^k d_k$ is integral. Their Remark 11 records two additional observations for $3 \leq k \leq 1000$: negativity and the congruence $r_k \equiv \binom{2k}{k} \pmod{32}$. Their proved Theorem 17 is the weaker statement $k!r_k \in \mathbb{Z}$ [1]. The OEIS entry A321941 still describes integrality as conjectural in the entry inspected on 1 October 2026 [2]. Its theorem numbers refer to an earlier preprint; our numbers refer to the final 2019 journal article.

Theorem 1.1 (Integrality and the full mod-32 statement). *For the coefficients in (1.2),*

$$r_0 = 1, \quad r_k = 64^k d_k \in 2\mathbb{Z} \quad (k \geq 1). \quad (1.3)$$

Moreover,

$$r_k \equiv \binom{2k}{k} + 16\mathbf{1}_{\{1,2\}}(k) \pmod{32} \quad (k \geq 0). \quad (1.4)$$

Thus the integrality conjecture and the congruence part of the stronger observation are true at every index.

The exceptional corrections in (1.4) are necessary: $r_1 = -14$ and $r_2 = 86$, whereas the corresponding central binomial coefficients are 2 and 6.

Theorem 1.2 (Denominators). *The positive integers B for which $B^k d_k \in \mathbb{Z}$ for every $k \geq 0$ are exactly the multiples of 64. For every $k \geq 1$,*

$$\text{den}(d_k) \mid 2^{6k-1}. \quad (1.5)$$

If $s_2(k)$ denotes the number of ones in the binary expansion of k , then

$$\text{den}(d_k) = 2^{6k-s_2(k)} \quad \text{whenever } 1 \leq s_2(k) \leq 4. \quad (1.6)$$

In particular, (1.5) is attained whenever k is a power of two.

Theorem 1.1 is proved in Sections 4 and 6. Theorem 1.2 is proved in Section 7. We also construct an integral polynomial family in Section 5 and invert the asymptotic expansion in Section 8.

1.1 What is new, and what is not claimed

The contribution is an arithmetic mechanism for this published conjecture, not the invention of asymptotic series, Lagrange inversion, or products of solutions of second-order equations. The crucial change is to use a *quadratic* product identity carrying the Wronskian, rather than only a *linear* recurrence for the product. In the new equation, the unknown coefficient is multiplied by 2, not by its index. Evenness then makes the only division exact.

The proof also yields a polynomial deformation and congruences that are not asserted in the cited source. These are presented as results established in this draft; global literature priority has not been certified. A targeted search for A321941 and for the published conjecture found the original paper and OEIS entry, but no subsequent proof. A repository code search for A321941 returned no matches; this is not a proof of absence from every file in the repository.

The separate assertion $r_k < 0$ for all $k \geq 3$ remains open in this work. No convergence, optimal truncation theorem, or exponentially small completion of (1.2) is inferred from formal coefficient arithmetic.

[Write note, batch 86A, 4 October 2026.] The assertion $r_k < 0$ for every $k \geq 3$ is proved in Part II (Theorem 13.1), by a different, analytic method; the statements of this Part about it describe the work of 1 October 2026. Convergence, optimal truncation and exponentially small completions remain open.

1.2 Place in the ProveIt collection

[Write note, batch 73O1, 1 October 2026.] This report is filed in ProveIt’s research-report collection, in the directory

SetTheory/Cardinals/docs/reports/congruences-and-valuations/
a321941-asymptotic-coefficient-integrality.

Its README there is the guide to the files.

Provenance. The report is built from one manuscript: manuscript 45 of batch 73 (cluster O1) of ProveIt’s incoming-reports intake, the archive A321941_Arithmetic_Conjectures.zip, which arrived in commit c79d64038 and was placed by commit 9df4ba51a. Its pin is the commit 0a6d798c5f39775b2fb4d85fef643fef996c87cf quoted in Section 10. The text is printed as delivered. The write changed only the label names, which now carry the prefix bgg:, and added the notes marked [Write note]. Nothing was merged into it; no other manuscript of the batch treats this sequence.

Status. The manuscript’s author line is “ChatGPT”. It is unrefereed, and nothing in it is formalized in Lean, Rocq or any other proof assistant. The intake reran the shipped verifier on a copy (exit code 0; the coefficient and numerical tables byte-identical to the shipped ones). Independently of the delivered code, it derived the symmetric-square recurrence of ρ_n from the recurrence of a_n , solved it order by order for $r_0, \dots, r_{10}$ (identical to the shipped table), and confirmed the residues modulo 32 for $k \leq 10$ and $\text{den}(d_k) = 2^{6k-s_2(k)}$ for $k \leq 8$. It did not re-derive every proof. The statements about Brent–Glasser–Guttmann’s paper (the range $3 \leq k \leq 1000$ of their Remark 11, their theorem numbers) and about the OEIS entry (its theorem numbering, and integrality still described as conjectural) are as the manuscript inspected them on 1 October 2026; the intake did not re-check them.

Shipped layout. The verifier is code/verify.py; its recorded outputs, the console log and the build receipt are in data/. The paths printed in Section 9 and Appendix B are the delivered ones; the delivered README and PDF are not shipped.

Formal neighbour. The Lean module Analysis/FabiusFunction/Lean/FabiusFunction

`n/QuadraticCoreCatalan.lean`, cited in Section 10 and unchanged since the pin, formalizes the Catalan closed form of $\binom{1/2}{n}$ (`Fabius.quadHalf_rat`) and the Catalan convolution (`Fabius.quadHalf_antidiagonal`). These are an ingredient of the fixed-point formulation (4.8), not of any theorem of this report as stated; the power-series identity itself is not formalized there, and this report gains no formal status from the module.

Neighbouring reports. In the same collection, `secant-number-periodicity` (OEIS A000364 modulo m) and `iterated-series/compositional-tree-series-congruences` (integrality and congruences of a formal fixed point) are arithmetic studies of other sequences; there is no overlap. The negativity assertion $r_k < 0$ for $k \geq 3$ remains open (question 1 of Section 11).

[Write note, batch 86A, 4 October 2026.] The previous sentence is superseded: Part II proves $r_k < 0$ for every $k \geq 3$, with the large-order law of r_k ; its provenance, status and notation are in Section 13.1.

2 The recurrence and its quadratic product invariant

2.1 A unit-determinant normalization

Put $G = eE_1(1)$. Direct differentiation of (1.1) gives

$$(1-z)^2 f'_0(z) - f_0(z) = 0, \quad (1-z)^2 f'_1(z) - f_1(z) = z - 1. \quad (2.1)$$

For the second identity, use $E'_1(x) = -e^{-x}/x$ and $x = (1-z)^{-1}$. Coefficient extraction yields

$$(n+1)u_{n+1} - (2n+1)u_n + (n-1)u_{n-1} = 0 \quad (n \geq 2) \quad (2.2)$$

for both $u = a$ and $u = b$. The initial values are

$$a_1 = 1, \quad a_2 = \frac{3}{2}, \quad b_1 = G - 1, \quad b_2 = \frac{3G - 2}{2}.$$

Set

$$x_n = na_n, \quad y_n = -nb_n, \quad p_n = x_n y_n = -n^2 \rho_n. \quad (2.3)$$

Then both normalized solutions satisfy

$$u_{n+1} = q_n u_n - u_{n-1}, \quad q_n = 2 + \frac{1}{n} \quad (n \geq 2). \quad (2.4)$$

Their discrete Wronskian

$$W_n = x_n y_{n+1} - x_{n+1} y_n$$

is constant: substituting (2.4) shows $W_n = W_{n-1}$. The initial values give

$$W_1 = 1(2 - 3G) - 3(1 - G) = -1.$$

The exact value $W_n^2 = 1$ is the normalization needed below.

For completeness, $p_n > 0$ for $n \geq 1$. The coefficients a_n are positive. For the other factor, the integral formula

$$b_n = - \int_0^1 e^{-s/(1-s)} s^{n-1} ds \quad (n \geq 1) \quad (2.5)$$

shows $b_n < 0$. One derivation starts from $e^x E_1(x) = \int_0^\infty e^{-u}(x+u)^{-1} du$, extracts the coefficient of z^n at $x = (1-z)^{-1}$, and substitutes $s = u/(1+u)$.

2.2 A general algebraic identity

Lemma 2.1 (Quadratic invariant). *Suppose $x_{n+1} = q_n x_n - x_{n-1}$ and $y_{n+1} = q_n y_n - y_{n-1}$. Write $p_n = x_n y_n$ and $W = x_n y_{n+1} - x_{n+1} y_n$. Then*

$$q_n^4 p_n^2 - 2q_n^2 p_n (p_{n+1} + p_{n-1}) + (p_{n+1} - p_{n-1})^2 = q_n^2 W^2. \quad (2.6)$$

This is a polynomial identity; no division by p_n or q_n is required.

Proof. Define $C_n = x_{n+1} y_n + x_n y_{n+1}$. The difference-of-squares identity gives

$$C_n^2 - 4p_n p_{n+1} = W^2.$$

Using $x_{n-1} = q_n x_n - x_{n+1}$ and the analogous identity for y gives

$$p_{n-1} = q_n^2 p_n - q_n C_n + p_{n+1},$$

so $q_n C_n = q_n^2 p_n + p_{n+1} - p_{n-1}$. Multiply the first equation by q_n^2 and substitute this expression. Expanding and collecting terms proves (2.6). $\square$

The Wronskian condition supplies the inhomogeneous constant on the right. Discarding it would leave an unnecessarily ambiguous formal product equation. In our application, (2.6) holds for every integer $n \geq 2$, with $q_n = 2 + 1/n$ and $W^2 = 1$.

3 From the exact recurrence to a formal master equation

Let

$$D(h) = \sum_{k \geq 0} d_k h^k \in \mathbb{Q}[[h]], \quad D(0) = 1. \quad (3.1)$$

This is a formal series: we do not evaluate an infinite convergent sum at $h = 1/n$. Equation (1.2) says that $p_n \sim \frac{1}{2}\sqrt{n} D(1/n)$ in the all-orders sense.

For any formal series D , define the two shifted expressions

$$U_{\pm}(h) = (1 \pm h)^{1/2} D\left(\frac{h}{1 \pm h}\right). \quad (3.2)$$

They encode $p_{n \pm 1}$ after the common factor $\frac{1}{2}\sqrt{n}$ is removed. Substitution into (2.6) gives, with $q = 2 + h$,

$$q^2 D^2 - 2D(U_+ + U_-) + \frac{(U_+ - U_-)^2}{q^2} = 4h. \quad (3.3)$$

Lemma 3.1 (Justification of the formal identity). *The coefficients from the published expansion satisfy (3.3) as an identity in $\mathbb{Q}[[h]]$.*

Proof. Fix an arbitrary finite target order. Expand p_n and $p_{n \pm 1}$ to sufficiently many additional orders in the exact identity (2.6). Shifts by a fixed integer preserve Poincaré expansions; the substitution $1/(n \pm 1) = h/(1 \pm h)$ and the binomial expansion of $(1 \pm h)^{1/2}$ give precisely (3.2). Products and division by the unit $q^2 = (2 + h)^2$ preserve finite-order expansions. After removing the common factor $n/4$, the exact identity gives (3.3) through the chosen order. A first nonzero formal coefficient would contradict this conclusion at its order. Since the target order was arbitrary, every coefficient agrees. $\square$

The even and odd parts of the shift make cancellation explicit. Put

$$E(D) = \frac{U_+ + U_- - 2D}{h^2}, \quad O(D) = \frac{U_+ - U_-}{h}. \quad (3.4)$$

Both quotients are formal series, because the indicated numerators have the required orders. Since

$$U_{\pm} = \sum_{j \geq 0} d_j h^j (1 \pm h)^{1/2-j},$$

their coefficients can be written explicitly using half-integral binomial coefficients. Inserting (3.4) into (3.3), and using $q^2 - 4 = 4h + h^2$, gives

$$D^2 = 1 - \frac{h}{4} D^2 + \frac{h}{2} D E(D) - \frac{h}{4(2+h)^2} O(D)^2. \quad (3.5)$$

The cancellation of one factor h is legitimate in $\mathbb{Q}[[h]]$.

Now scale

$$R(z) = D(64z) = \sum_{k \geq 0} r_k z^k. \quad (3.6)$$

For $j, \ell \geq 0$, define rational weights

$$e_{j,\ell} = 8 \cdot 64^{2\ell} \binom{1/2-j}{2\ell+2}, \quad o_{j,\ell} = 2 \cdot 64^{2\ell} \binom{1/2-j}{2\ell+1}. \quad (3.7)$$

For a series $F(z) = \sum_{j \geq 0} f_j z^j$, set

$$\mathcal{E}F = \sum_{j,\ell \geq 0} e_{j,\ell} f_j z^{j+2\ell}, \quad \mathcal{O}F = \sum_{j,\ell \geq 0} o_{j,\ell} f_j z^{j+2\ell}. \quad (3.8)$$

Each coefficient is a finite sum. Directly from the definitions, $\mathcal{E}R = 4E(D)(64z)$ and $\mathcal{O}R = O(D)(64z)$.

Proposition 3.2 (The scaled master equation). *The series R satisfies*

$$\boxed{R^2 = 1 + 4z \left(-4R^2 + 2R\mathcal{E}R - \frac{(\mathcal{O}R)^2}{(1+32z)^2} \right)}. \quad (3.9)$$

Conversely, among formal series in $\mathbb{Q}[[z]]$ with constant term 1, (3.9) has at most one solution.

Proof. The identity is the substitution $h = 64z$ in (3.5). For uniqueness, the coefficient of z^k on the left is $2r_k$ plus an expression in earlier coefficients. The coefficient of z^k on the right depends only on coefficients through r_{k-1} : there is an external factor z , and the operators never decrease degree. Therefore r_k is determined recursively. $\square$

4 Integral shift operators and the integrality proof

4.1 The exact powers of two in the shift weights

Lemma 4.1 (Half-binomial arithmetic). *For integers $j \geq 0$ and $m \geq 1$, the number $\binom{1/2-j}{m}$ is dyadic, and*

$$v_2 \binom{1/2-j}{m} = -m - v_2(m!) = -2m + s_2(m). \quad (4.1)$$

Consequently, all weights in (3.7) are integers. More precisely,

$$e_{j,0} = 4j^2 - 1, \quad o_{j,0} = 1 - 2j, \quad (4.2)$$

$$256 \mid e_{j,\ell}, \quad 512 \mid o_{j,\ell} \quad (\ell \geq 1). \quad (4.3)$$

Proof. The elementary formula

$$\binom{1/2}{m} = \frac{(-1)^{m-1} \text{Cat}_{m-1}}{2^{2m-1}} \quad (m \geq 1), \quad \text{Cat}_r = \frac{1}{r+1} \binom{2r}{r} \in \mathbb{Z},$$

shows that the coefficients of $(1+x)^{1/2}$ are dyadic. Multiplication by $(1+x)^{-j} \in \mathbb{Z}[[x]]$ proves the same for $\binom{1/2-j}{m}$. Alternatively, this is the finite Vandermonde convolution. In the product

$$\binom{1/2-j}{m} = \frac{\prod_{i=0}^{m-1} (1-2j-2i)}{2^m m!},$$

the numerator is odd. This proves the first equality in (4.1). The identity $v_2(m!) = m - s_2(m)$ gives the second.

For the even and odd weights, respectively, substitution yields

$$v_2(e_{j,\ell}) = 8\ell - 1 + s_2(\ell + 1), \quad v_2(o_{j,\ell}) = 8\ell + s_2(\ell). \quad (4.4)$$

For $\ell = 0$, both valuations are zero, and direct calculation gives (4.2). For $\ell \geq 1$, the valuations are at least 8 and 9, respectively. Dyadicity together with these nonnegative valuations proves integrality, not merely the absence of a factor 2 in a denominator. $\square$

Writing $\theta = z \frac{d}{dz}$, the lemma implies, for every $F \in \mathbb{Z}[[z]]$,

$$\mathcal{E}F - (4\theta^2 - 1)F \in 256z^2\mathbb{Z}[[z]], \quad \mathcal{O}F - (1 - 2\theta)F \in 512z^2\mathbb{Z}[[z]]. \quad (4.5)$$

These stronger divisibilities will be used for the congruence theorem.

4.2 The triangular recurrence

Define

$$\mathcal{H}(F) = -4F^2 + 2F\mathcal{E}F - \left(\frac{\mathcal{O}F}{1+32z} \right)^2. \quad (4.6)$$

Lemma 4.1 shows that $\mathcal{H}$ maps integral series to integral series. Let $R_{<k} = \sum_{j=0}^{k-1} r_j z^j$. Extracting z^k from (3.9) gives

$$r_k = 2[z^{k-1}]\mathcal{H}(R_{<k}) - \frac{1}{2} \sum_{j=1}^{k-1} r_j r_{k-j}. \quad (4.7)$$

This expression depends only on already determined coefficients.

Proof of the integrality assertion in Theorem 1.1. Start with $r_0 = 1$. Suppose $r_j \in 2\mathbb{Z}$ for $1 \leq j < k$. Then $[z^{k-1}]\mathcal{H}(R_{<k})$ is an integer. Every summand in the convolution in (4.7) is divisible by 4, so half of the whole sum is an even integer. Both terms on the right are therefore even integers. This proves $r_k \in 2\mathbb{Z}$ by induction. Proposition 3.2 and Lemma 3.1 identify this recursively constructed series with the coefficients of the analytic expansion. $\square$

The original index division has disappeared. This is why the quadratic identity proves more than the published linear recurrence, whose direct arithmetic consequence was $k!r_k \in \mathbb{Z}$. The proof establishes actual integrality of r_k ; it does not assume the finite OEIS values are integers and extrapolate.

The first steps illustrate the mechanism. At constant order, $\mathcal{E}1 = -1 + O(z^2)$ and $\mathcal{O}1 = 1 + O(z^2)$, so $\mathcal{H}(1) = -7 + O(z)$. Thus $r_1 = 2(-7) = -14$. Subsequent applications give $r_2 = 86$, $r_3 = -3660$, and the table in Section 9.

4.3 A Catalan fixed-point interpretation

There is also a useful division-free formulation. The formal identity

$$\sqrt{1 + 4X} = 1 + 2 \sum_{m \geq 1} (-1)^{m-1} \text{Cat}_{m-1} X^m \quad (4.8)$$

has integer coefficients. Thus the map

$$\Phi(F) = 1 + 2 \sum_{m \geq 1} (-1)^{m-1} \text{Cat}_{m-1} (z\mathcal{H}(F))^m$$

maps $\mathbb{Z}[[z]]$ into $1 + 2z\mathbb{Z}[[z]]$. Because the operators in $\mathcal{H}$ do not decrease degree, agreement of F and G through degree $k - 1$ implies agreement of $\Phi(F)$ and $\Phi(G)$ through degree k . Successive substitution therefore determines a unique formal fixed point. Over $\mathbb{Q}[[z]]$, squaring identifies its equation with (3.9).

This interpretation explains why Catalan arithmetic is relevant. It also suggests a particularly clean formalization route: construct the fixed point by finite coefficient truncations and use the Catalan convolution, rather than formal square-root analysis.

5 An integral polynomial deformation

The same proof works with an indeterminate parameter. This is more than an integer sequence coincidence: the coefficient arithmetic survives a one-parameter deformation of the discrete equation.

5.1 Definition and integrality

Let t be an indeterminate. For $F = \sum f_j(t)z^j$, define

$$\mathcal{E}_t F = \sum_{j, \ell \geq 0} e_{j, \ell} t^\ell f_j(t) z^{j+2\ell}, \quad \mathcal{O}_t F = \sum_{j, \ell \geq 0} o_{j, \ell} t^\ell f_j(t) z^{j+2\ell}. \quad (5.1)$$

These are well-defined, degree-nondecreasing linear operators on $\mathbb{Z}[t][[z]]$. Put

$$\mathcal{H}_t(F) = -4tF^2 + 2F\mathcal{E}_t F - \left(\frac{\mathcal{O}_t F}{1 + 32tz} \right)^2. \quad (5.2)$$

Theorem 5.1 (Polynomial master series). *There is a unique series $R_t \in 1 + z\mathbb{Q}[t][[z]]$ satisfying*

$$R_t^2 = 1 + 4z\mathcal{H}_t(R_t). \quad (5.3)$$

Writing $R_t(z) = \sum_{k \geq 0} P_k(t)z^k$, one has

$$P_0(t) = 1, \quad P_k(t) \in 2\mathbb{Z}[t] \quad (k \geq 1), \quad P_k(1) = r_k. \quad (5.4)$$

Every P_k has degree exactly k , with

$$[t^k]P_k(t) = (-4)^k \binom{2k}{k}. \quad (5.5)$$

Proof. The recurrence (4.7), with $\mathcal{H}$ replaced by $\mathcal{H}_t$, determines each coefficient from its predecessors. The evenness induction works coefficientwise over $\mathbb{Z}[t]$ and proves existence and (5.4). Uniqueness over $\mathbb{Q}[t]$ follows from the coefficient $2P_k$ on the left. At $t = 1$, the equation is (3.9), proving $P_k(1) = r_k$.

Inductively, assume $\deg P_j \leq j$. In the coefficient of z^m in $\mathcal{E}_t R_t$ or $\mathcal{O}_t R_t$, a summand with $j + 2\ell = m$ has degree at most $j + \ell \leq m$. The same degree bound is preserved by multiplication and by $(1 + 32tz)^{-1}$. Consequently, the recurrence gives $\deg P_k \leq k$.

In total bidegree $t^k z^k$, the terms $8zR_t \mathcal{E}_t R_t$ and $-4z(\mathcal{O}_t R_t / (1 + 32tz))^2$ have insufficient t -degree. If $S(w) = \sum_{k \geq 0} [t^k] P_k(t) w^k$, the diagonal terms of (5.3) therefore satisfy

$$S(w)^2 = 1 - 16wS(w)^2.$$

The branch with constant term 1 is $S(w) = (1 + 16w)^{-1/2}$, whose k th coefficient is $(-4)^k \binom{2k}{k} \neq 0$. This proves both the degree and (5.5). $\square$

The first polynomials are

$$\begin{aligned} P_1(t) &= -8t - 6, \\ P_2(t) &= 96t^2 + 80t - 90, \\ P_3(t) &= -1280t^3 + 2240t^2 + 1680t - 6300, \\ P_4(t) &= 17920t^4 - 161280t^3 - 57792t^2 + 151200t - 992250. \end{aligned}$$

The alternating and mixed signs already show why integrality alone does not prove the separate negativity assertion at $t = 1$.

5.2 The constant edge has a closed form

Theorem 5.2 (Constant coefficients). *For every $k \geq 1$,*

$$P_k(0) = -\binom{2k}{k} (2k-3)!!(2k+1)!!, \quad (5.6)$$

where $(-1)!! = 1$. In particular, this edge of the polynomial family is strictly negative at every positive index.

Proof. Set $S(z) = R_0(z)$. At $t = 0$, only the diagonal operator weights survive: $\mathcal{E}_0 S = (4\theta^2 - 1)S$ and $\mathcal{O}_0 S = (1 - 2\theta)S$. Hence

$$S^2 = 1 + 4z(-3S^2 + 4S\theta S + 8S\theta^2 S - 4(\theta S)^2). \quad (5.7)$$

Apply θ to both sides. If the parenthesized expression is denoted by F , its derivative satisfies

$$F + \theta F = S(-3S - 2\theta S + 12\theta^2 S + 8\theta^3 S).$$

Cancel the unit S to obtain

$$\theta S = 2z(2\theta + 3)(2\theta - 1)(2\theta + 1)S. \quad (5.8)$$

Coefficient extraction gives

$$kP_k(0) = 2(2k+1)(2k-3)(2k-1)P_{k-1}(0). \quad (5.9)$$

The expression on the right of (5.6) has initial value -6 at $k = 1$ and the same consecutive ratio. Since $P_0(0) = 1$, (5.9) proves the formula by induction. $\square$

Thus the extreme coefficients of each polynomial are explicit, although the interior coefficients need not have one sign. Formula (5.8) is a useful linear equation at this special parameter, not a replacement for the nonlinear integrality argument at $t = 1$.

[Write note, batch 86A, 4 October 2026.] The negative edge $P_k(0) = -S_k$ is the leading asymptotic scale of $P_k(1) = r_k$: Theorem 13.2 of Part II gives $r_k/P_k(0) = 1 - 1/(3k) + O(k^{-2})$, with an expansion to every order.

5.3 Why this deformation is natural

Consider the recurrence $u_{n+1} = (2 + \lambda/n)u_n - u_{n-1}$ over a field containing a nonzero parameter λ and a square root of λ . Seek a formal solution of the quadratic identity, with Wronskian square 1, of the form

$$p_n \sim \frac{\sqrt{n}}{2\sqrt{\lambda}} D_\lambda(1/n), \quad D_\lambda(0) = 1.$$

Here “formal product” means the normalized solution of the quadratic identity; analytic existence for all parameters is not assumed. Repeating the calculation of Section 3 gives

$$D_\lambda^2 = 1 - \frac{\lambda h}{4} D_\lambda^2 + \frac{h}{2\lambda} D_\lambda E(D_\lambda) - \frac{h}{4\lambda(2 + \lambda h)^2} O(D_\lambda)^2. \quad (5.10)$$

Set $h = 64\lambda z$ and $t = \lambda^2$. The even and odd shift weights then become exactly (5.1), and (5.10) becomes (5.3). Therefore

$$[h^k] D_\lambda(h) = \frac{P_k(\lambda^2)}{(64\lambda)^k}. \quad (5.11)$$

This is an unconditional statement about the unique normalized formal solution. Whenever an actual normalized product has an all-orders expansion, the formal-bridge argument identifies its coefficients with (5.11). The case $\lambda = 1$ is established by the published analytic theorem. The specialization $t = 0$ in Theorem 5.2 is a formal polynomial specialization; it is not a claimed uniform analytic limit as $\lambda \rightarrow 0$.

6 The congruence modulo 32

The large off-diagonal powers of two in Lemma 4.1 allow a short reduction of the entire master equation. We prove the stronger polynomial statement first.

Theorem 6.1 (Polynomial congruence). *For every $k \geq 0$, coefficientwise in $\mathbb{Z}[t]$,*

$$P_k(t) \equiv (-1)^k (4t + 3)^k \binom{2k}{k} + 16 \mathbf{1}_{\{2\}}(k) \pmod{32}. \quad (6.1)$$

Equivalently,

$$R_t(z) \equiv (1 + (16t + 12)z)^{-1/2} + 16z^2 \pmod{32}. \quad (6.2)$$

Proof. Write $R = R_t$ temporarily. The operator congruences (4.5) remain valid over $\mathbb{Z}[t]$, and $(1 + 32tz)^{-2} \equiv 1 \pmod{64}$. Reducing (5.3) modulo 64 gives

$$R^2 \equiv 1 + 4z(-(4t + 3)R^2 + 4R\theta R + 8R\theta^2 R - 4(\theta R)^2) \pmod{64}. \quad (6.3)$$

Theorem 5.1 shows that both θR and $\theta^2 R$ are even series. Hence the last two terms, after multiplication by $4z$, vanish modulo 64. With $\alpha = 16t + 12$, we obtain

$$(1 + \alpha z)R^2 \equiv 1 + 16zR\theta R \pmod{64}. \quad (6.4)$$

Let $C = (1 + \alpha z)^{-1/2}$. Its coefficients are $(-1)^k (4t + 3)^k \binom{2k}{k}$, so $C \in 1 + 2z\mathbb{Z}[t][[z]]$. Define $S = C + 16z^2$. Since $C \equiv 1 \pmod{2}$ and $\alpha \equiv 0 \pmod{4}$,

$$\begin{aligned} (1 + \alpha z)S^2 &\equiv 1 + 32z^2 \pmod{64}, \\ 16zS\theta S &\equiv 16zC\theta C = -\frac{8\alpha z^2}{(1 + \alpha z)^2} \equiv 32z^2 \pmod{64}. \end{aligned}$$

For the last congruence, $-8\alpha \equiv 32 \pmod{64}$, and $(1 + \alpha z)^{-2} \equiv 1 \pmod{2}$. Thus S satisfies the same reduced equation as R .

We now justify the loss of exactly one power of two when taking the square root; no division by 2 in $\mathbb{Z}/64\mathbb{Z}$ is being assumed. Both R and S have constant term 1 and even higher coefficients. Suppose their coefficients below degree k agree modulo 32. In degree k , $R^2 - S^2$ is $2([z^k]R - [z^k]S)$ modulo 64: every remaining product contains an even nonconstant coefficient and a difference divisible by 32. The term $\alpha z(R^2 - S^2)$ and the difference $16z(R\theta R - S\theta S)$ depend only on lower degrees and vanish modulo 64. Subtracting their two instances of (6.4) therefore gives

$$2([z^k]R - [z^k]S) \equiv 0 \pmod{64}.$$

Thus $[z^k]R \equiv [z^k]S \pmod{32}$. Induction proves (6.2), and coefficient extraction gives (6.1). $\square$

Proof of the congruence assertion in Theorem 1.1. Specialize (6.2) to $t = 1$:

$$R(z) \equiv (1 + 28z)^{-1/2} + 16z^2 \pmod{32}.$$

Factor $1 + 28z = (1 - 4z)(1 + 32z/(1 - 4z))$. The binomial expansion gives

$$(1 + 28z)^{-1/2} \equiv (1 - 4z)^{-1/2} - 16z(1 - 4z)^{-3/2} \pmod{32}.$$

All terms of second and higher degree in $32z/(1 - 4z)$ vanish modulo 32; this follows directly from their half-binomial valuations. The nonconstant coefficients of $(1 - 4z)^{-3/2}$ are even: its k th coefficient is $(2k + 1)\binom{2k}{k}$. Consequently,

$$R(z) \equiv (1 - 4z)^{-1/2} + 16z + 16z^2 \pmod{32}. \quad (6.5)$$

Using $[z^k](1 - 4z)^{-1/2} = \binom{2k}{k}$ proves (1.4) at all indices, including the exceptions. $\square$

The proof demonstrates why a central binomial sequence appears: modulo a small power of two, the nonlocal shift equation collapses to a differential quadratic equation whose solution is an inverse square root, up to two low-degree corrections.

7 Minimal scaling and exact denominators

Proof of Theorem 1.2. Since r_k is even for $k \geq 1$, the denominator of $d_k = r_k/2^{6k}$ divides 2^{6k-1} . The computed value

$$d_2 = \frac{43}{2048} = \frac{43}{2^{11}}$$

shows that $B^2 d_2 \in \mathbb{Z}$ forces $2v_2(B) \geq 11$, hence $v_2(B) \geq 6$. Thus every permissible positive B is divisible by 64. Conversely, $B = 64m$ gives $B^k d_k = m^k r_k \in \mathbb{Z}$, proving the exact classification of scales.

The factorial identity $v_2(k!) = k - s_2(k)$ gives

$$v_2\binom{2k}{k} = (2k - s_2(2k)) - 2(k - s_2(k)) = s_2(k). \quad (7.1)$$

For $k \geq 3$ with $s_2(k) \leq 4$, the congruence $r_k \equiv \binom{2k}{k} \pmod{32}$ forces $v_2(r_k) = s_2(k)$. Indeed, adding a multiple of 2^5 cannot change a valuation strictly less than 5. The indices $k = 1, 2$ are checked directly: $r_1 = -14$ and $r_2 = 86$ both have valuation $1 = s_2(k)$. Cancelling the exact power of two in $d_k = r_k/2^{6k}$ proves (1.6). For $k = 2^m$, $s_2(k) = 1$, so the bound is attained at every such index. $\square$

This separates two notions of optimality. The base 64 is the smallest integer exponential scale making every coefficient integral. The additional factor 2 available at every positive index improves the denominator bound, but cannot reduce that exponential base. The equality at infinitely many powers of two shows that the improved bound is not merely an artifact of the proof.

For indices with at least five binary ones, the mod-32 congruence only forces $32 \mid r_k$ and does not determine the exact valuation. In particular, we do not extrapolate (1.6) to all indices.

8 All-orders inversion of the product expansion

The arithmetic series also supports an explicit inverse expansion. It is important to specify what is being inverted: a continuous global inverse of an interpolation of the sequence has not been chosen.

Define the positive real sequence

$$y_n = (-2\rho_n)^{-2/3}. \quad (8.1)$$

Positivity follows from (2.5). The known product expansion gives

$$y_n \sim nD(1/n)^{-2/3}, \quad y_n \sim n. \quad (8.2)$$

Theorem 8.1 (Inverse expansion). *Define*

$$\beta_0 = \frac{2}{3}d_1, \quad \beta_k = -\frac{1}{k}[h^{k+1}]D(h)^{-2k/3} \quad (k \geq 1). \quad (8.3)$$

Then, for each fixed $M \geq 0$,

$$n = y_n + \beta_0 + \sum_{k=1}^M \beta_k y_n^{-k} + O(y_n^{-M-1}) \quad (n \rightarrow \infty). \quad (8.4)$$

In particular,

$$\begin{aligned} n = y_n - \frac{7}{48} - \frac{29}{2304y_n} - \frac{3719}{331776y_n^2} - \frac{1565}{32768y_n^3} \\ - \frac{36355159}{191102976y_n^4} + O(y_n^{-5}). \end{aligned} \quad (8.5)$$

Moreover,

$$64^{k+1}\beta_k \in \mathbb{Z}[1/3] \quad (k \geq 0). \quad (8.6)$$

Thus every inverse coefficient has denominator supported only at the primes 2 and 3, and after the displayed normalization only 3 remains.

Proof. Let $h = 1/n$ and formally put $u = 1/y$. The forward relation becomes

$$u = hD(h)^{2/3}, \quad h = u\psi(h), \quad \psi(h) = D(h)^{-2/3}.$$

The series $u = h + O(h^2)$ has a unique formal compositional inverse. For $k \geq 1$, Lagrange inversion applied to the Laurent function h^{-1} gives

$$[u^k] \frac{1}{h(u)} = -\frac{1}{k}[h^{k+1}]\psi(h)^k,$$

which is (8.3). This Laurent version follows, for example, by changing variables in the coefficient residue and integrating a formal derivative; it has the same finite coefficient meaning as the usual

power-series version. The constant term is obtained separately from $h = u + \psi'(0)u^2 + O(u^3)$: it is $-\psi'(0) = 2d_1/3$.

For the asymptotic assertion, use the published expansion to enough fixed orders and form (8.2) by finite-order Taylor expansion around 1. Substitute the truncated forward expansion into the formal inverse. All coefficients through y^{-M} cancel by construction. Since $y_n \sim n$, the uncanceled terms and transported remainders are $O(y_n^{-M-1})$. No global invertibility or monotonicity assumption is needed for this composition along the sequence. Extracting the first coefficients gives (8.5).

For arithmetic, introduce $X = 64n$, $Y = 64y$ and $z = 1/X$. The normalized forward relation is

$$Y = XR(1/X)^{-2/3}, \quad Y^{-1} = zR(z)^{2/3}.$$

Because $R \in \mathbb{Z}[[z]]$, the unit $F = R^{2/3}$ belongs to $\mathbb{Z}[1/3][[z]]$: solve $F^3 = R^2$ coefficient by coefficient, dividing only by 3. A monic series $zF(z) = z + O(z^2)$ has a compositional inverse over the same ring, and reciprocating that inverse gives

$$X(Y) \in Y + \mathbb{Z}[1/3][[Y^{-1}]].$$

Since the coefficient of Y^{-k} in $X(Y) - Y$ is $64^{k+1}\beta_k$, this proves (8.6). $\square$

8.1 The boundary between an expansion and a transseries

Theorem 8.1 gives the complete algebraic sector of an inverse asymptotic description. It does not specify exponentially small corrections. Two functions differing by a term smaller than every inverse power have the same formal series, so coefficient arithmetic cannot recover such a term. This distinction is especially relevant here: the individual factors have opposite stretched-exponential behavior, while their product has an ordinary inverse-power expansion.

Similarly, an $O(y_n^{-M-1})$ statement supplies no numerical error constant at a specified n . Rigorous rounding of an inverse estimate to an integer would require an effective remainder bound. The numerical checks below illustrate the expansion but are not interval error certificates.

9 Reproducible exact and numerical checks

[Write note, batch 7301.] In the collection the program is `code/verify.py` and the files it writes are shipped in `data/`. Its default output directory is its own, `code/`; pass `-out` to write elsewhere (Appendix B).

The accompanying program `verify.py` uses two different coefficient recurrences. The new method computes r_k using integer arithmetic and the quadratic master equation. The independent method uses rational arithmetic and the published linear coefficient recurrence. Agreement is a useful check on normalization, signs, and indexing; the all-orders proofs above do not depend on how far the calculations are run.

9.1 An explicit implementation form

Let

$$R = \sum r_n z^n, \quad \mathcal{E}_t R = \sum E_n z^n, \quad \mathcal{O}_t R = \sum O_n z^n, \quad \frac{\mathcal{O}_t R}{1 + 32tz} = \sum T_n z^n.$$

Starting with $r_0 = 1$, $E_0 = -1$, $O_0 = T_0 = 1$, the integer recurrence is

$$H_{n-1} = -4t \sum_{i=0}^{n-1} r_i r_{n-1-i} + 2 \sum_{i=0}^{n-1} r_i E_{n-1-i} - \sum_{i=0}^{n-1} T_i T_{n-1-i}, \quad (9.1)$$

$$r_n = 2H_{n-1} - \frac{1}{2} \sum_{i=1}^{n-1} r_i r_{n-i}, \quad (9.2)$$

$$E_n = \sum_{\ell=0}^{\lfloor n/2 \rfloor} e_{n-2\ell, \ell} t^\ell r_{n-2\ell}, \quad O_n = \sum_{\ell=0}^{\lfloor n/2 \rfloor} o_{n-2\ell, \ell} t^\ell r_{n-2\ell}, \quad (9.3)$$

$$T_n = O_n - 32tT_{n-1}. \quad (9.4)$$

The order matters: compute r_n from earlier arrays, then append E_n, O_n, T_n . The division by two in (9.2) is always exact, and its quotient is even. The code asserts this at each step.

For comparison, the independent recurrence from [1, Lemma 14] is

$$8kd_k = - \sum_{j=0}^{k-1} d_j [h^{k+2-j}] \left(\frac{B(h)}{(1-h)^{j+1/2}} + \frac{C(h)}{(1-2h)^{j+1/2}} + \frac{J(h)}{(1-3h)^{j+1/2}} \right), \quad (9.5)$$

where

$$\begin{aligned} B(h) &= -6 + 13h - 7h^2 + h^3, \\ C(h) &= 6 - 19h + 17h^2 - 3h^3, \\ J(h) &= -2 + 11h - 17h^2 + 6h^3. \end{aligned}$$

The source calls the last polynomial D ; we rename it J to avoid collision with the asymptotic series. This second implementation does not use (9.2) or its integrality conclusion.

9.2 Initial coefficients

Table 1: Exact initial coefficients. The normalized product is $-2n^{3/2}\rho_n \sim \sum d_k n^{-k}$.

k	r_k	d_k in lowest terms
0	1	1
1	-14	-7/32
2	86	43/2048
3	-3660	-915/65536
4	-1042202	-521101/8388608
5	-247948260	-61987065/268435456
6	-108448540420	-27112135105/17179869184

The archived run used Python 3.13.5, SymPy 1.14.0, and mpmath 1.3.0. It verified integrality/evenness, the mod-32 formula, and all applicable valuation formulas through $k = 256$; independent agreement with (9.5) through $k = 128$; all thirteen displayed OEIS terms; the polynomial identities through degree 8; and the parameter congruence through $k = 48$ at $t = -3, -1, 0, 2, 5$. It also checked the bilinear invariant symbolically and composed the displayed inverse series directly.

The check that $r_k < 0$ for $3 \leq k \leq 256$ is explicitly logged as an *observation*, not a proof. It does not extend the range already reported in the 2019 paper and is not presented as a computational record. [Write note, batch 86A: the infinite assertion is proved in Part II, Theorem 13.1; this finite check remains an observation of this Part.]

9.3 Numerical asymptotic checks

For numerical evaluation we used the exact special-function expression

$$\rho_n = -e^{-1}\Gamma(n)M(n+1, 2, 1)U(n, 0, 1), \quad (9.6)$$

with the standard Kummer functions, as in [1]. This avoids computing the recessive b_n by unstable forward propagation from nearly cancelling initial data. Table 2 shows the signed errors from the published product and our inverse expansion.

Table 2: Numerical diagnostics, not rigorous remainder bounds. Product error: $-2n^{3/2}\rho_n - \sum_{k=0}^5 d_k n^{-k}$. Inverse error: the right side of (8.5), without its O term, minus n .

n	Product error	Inverse error
25	$-1.16668669188 \times 10^{-8}$	$2.14328374121 \times 10^{-7}$
100	$-1.75696329807 \times 10^{-12}$	$1.34214874476 \times 10^{-10}$
400	$-3.95020779621 \times 10^{-16}$	$1.21710421134 \times 10^{-13}$

The recorded numerical strings were unchanged when working precision was increased from 80 to 120 decimal digits. This tests numerical stability of the reported digits but is not interval arithmetic. Exact assertions are separate from these numerical diagnostics.

To reproduce the archived run, execute

```
python verify.py --order 256 --independent-order 128
```

The main exact checks require only the Python standard library. Installing SymPy and mpmath enables the additional checks; missing optional packages produce explicit skip messages. The files `verification_output.txt` and `verification_metadata.json` record what actually ran.

10 Connection with ProveIt and a formalization plan

The repository inspection was pinned to commit `0a6d798c5f39775b2fb4d85fef643fef996c87cf`. The canonical transseries-and-inversion README describes formal coefficient uniqueness, quadratic/Catalan coefficients, remainder transport, and the distinction between algebraic expansions and flat corrections [4]. The source file `QuadraticCoreCatalan.lean` was also inspected [5]. It contains the Catalan description of $\binom{1/2}{m}$ and finite convolution identities. It explicitly distinguishes those finite algebraic results from a complete formal-power-series existence/uniqueness package.

These are useful building blocks, not certificates for this article. In particular, no claim here relies on a repository statement about A321941, and no Lean or Rocq build was run for these new theorems.

A sensible formalization separates three layers. The first is finite algebra: prove the Wronskian identity over a commutative ring and the half-binomial weight formulas over $\mathbb{Q}$. The second is formal arithmetic: define the coefficient arrays over $\mathbb{Z}[t]$, prove evenness, establish uniqueness after coercion to $\mathbb{Q}[t]$, and formalize the modulo-64 argument that descends to modulo 32. The third is the analytic bridge: connect actual Kummer/exponential-integral coefficients to the formal product series by a full Poincaré expansion theorem.

The arithmetic layer can be formalized before the analytic one. Its theorem would say that the uniquely defined formal solution has integral even coefficients and the stated congruence. It should not be advertised as the full analytic A321941 theorem until the expansion-identification

bridge is also verified. This separation makes the remaining trusted assumptions visible rather than hiding them inside an informal numerical definition.

For implementation, (9.1)–(9.4) provide finite sums only. The first proof can use a parity invariant asserting that all stored coefficients except the constant term are even. The congruence proof needs a second invariant tracking equality modulo 32 while comparing squares modulo 64. The inspected Catalan module supports the alternative fixed-point construction in (4.8). The inverse arithmetic theorem requires only unit inversion, monic compositional inversion, and the recursion $F^3 = R^2$ over $\mathbb{Z}[1/3]$.

11 Further research questions and topics

The following questions are not asserted as solved. They are arranged from immediate extensions of the present arithmetic to broader analytic and structural problems.

- 1. Prove the remaining negativity conjecture.** Establish $P_k(1) < 0$ for all $k \geq 3$. The exact formula for $P_k(0)$ gives a negative reference edge, but mixed signs in the interior prevent a direct coefficientwise argument. A positive integral representation for $-r_k$, a sign-preserving reformulation, or an effective comparison with the constant edge would resolve the missing assertion in the published Remark 11.

[Write note, batch 86A, 4 October 2026: answered.] Part II proves $r_k < 0$ for every $k \geq 3$ (Theorem 13.1), not through the polynomials P_k but through an exact radial-oscillator heat kernel: r_k has the sign of a coefficient h_k dominated by the single negative term $-\gamma_k$ of (17.1) for $k \geq 32$, and an exact rational certificate covers $3 \leq k \leq 31$. The dominating term is the constant edge, $64^k(\frac{1}{2})_k \gamma_k = -P_k(0)$, so the proof is an effective comparison with the constant edge in the sense suggested here. The answer is AI-assisted and unrefereed.

- 2. Determine all exact 2-adic valuations.** Theorem 1.2 handles indices with at most four binary ones. Determine $v_2(r_k)$ at the remaining indices, including whether there is a rule in terms of binary carries and a finite amount of additional state. The existing proof gives a route through successively higher moduli, but no such complete rule has been established here.
- 3. Lift the universal congruence to higher powers of two.** Find an effective description of R_t modulo 2^m for arbitrary fixed m . The valuations in (4.4) imply that only finitely many shift distances are visible modulo each fixed power of two. Can this be organized into a finite differential system, a recursive correction to an algebraic series, or an efficient modular coefficient algorithm with proved complexity?
- 4. Study large-order growth of the coefficients themselves.** The expansion parameter is $n \rightarrow \infty$, but the coefficient index $k \rightarrow \infty$ is a different asymptotic problem. Determine a sharp growth law for r_k and the location and type of singularities of a suitable Borel transform. The exact edge (5.6) suggests a useful comparison scale, but does not by itself prove the growth of $P_k(1)$. Such a result would guide optimal truncation and could help prove eventual negativity, leaving only finitely many indices.

[Write note, batch 86A, 4 October 2026: answered in part.] Part II proves the sharp growth law $r_k \sim -16^k(k!)^2/(\pi^{3/2}k^{3/2})$, with a complete expansion in inverse powers of k relative to the edge $P_k(0)$ (Theorem 13.2), and it was used to prove negativity outright. For the order-two Borel transform $\sum_k d_k s^k/(2k)!$ it proves radius 16 and a logarithmic singularity at $s = 16$, as a radial boundary law (Theorem 19.1). The analytic continuation, the full singularity set and Stokes data stay open (Research question 21.2). The answer is AI-assisted and unrefereed.

- 5. Give effective analytic remainders and inverse certification.** Combine the arithmetic recurrences with explicit bounds for the product of Kummer functions. Determine an error bound after a specified truncation that is valid for all n above a computable threshold. Transport it through the inverse map to obtain certified integer reconstruction or rounding. The current all-orders theorem and high-precision checks do not supply these constants.
- 6. Analyze the parameter polynomials.** Investigate the real zeros, sign intervals, coefficient triangles, and large- k asymptotics of $P_k(t)$. Determine for which fixed real or integer t the signs eventually stabilize. Establish analytic realizations of (5.11) with uniform control in λ away from zero, and distinguish those from any singular scaling limit as $\lambda \rightarrow 0$.
- 7. Extend the arithmetic mechanism to other recurrences.** For normalized equations with $q_n = 2 + \lambda/n + \mu/n^2 + \dots$, classify the potentials and normalizations for which the diagonal-product equation admits integral coefficient operators after a single exponential scaling. The quadratic invariant survives unchanged; the challenge is the arithmetic of the shifted normalization and the resulting triangular equation. This is a systematic source of further OEIS-related integrality problems rather than a collection of isolated coefficient guesses.
- 8. Find a signed combinatorial interpretation.** The Catalan fixed point expresses the coefficients using trees built from integer-weighted operations, but this is not yet a natural combinatorial model for A321941. A model with a sign-reversing involution could explain both dyadic scaling and the conjectured negative sign. It should account for the parameter t and the central-binomial reduction, not merely encode an arbitrary integer recurrence tautologically.
- 9. Complete a machine-checked proof and an independently reviewed edition.** Formalize the finite arithmetic layer first, then the formal-analytic bridge. An independent mathematical review should focus on the normalized Wronskian, the cancellation leading to (3.5), the exact valuations of shift weights, and the induction from modulus 64 to modulus 32. A reviewed proof can then support a precise OEIS update distinguishing the settled arithmetic assertions from the remaining sign question.

12 Conclusion

The conjectural integrality of A321941 is not an accidental cancellation visible only in a long list of coefficients. It follows from an exact quadratic identity, integral rescaled shift operators, and a parity induction. The same identity reduces modulo 64 to a simple equation that proves the central-binomial congruence modulo 32.

The resulting structure is stronger than the original sequence statement: it gives even integral parameter polynomials, explicit extreme coefficients, minimal exponential scaling, infinitely many sharp denominator instances, and an arithmetic restriction on an all-orders inverse expansion. The proof is independent of finite verification ranges, while the accompanying checks provide reproducible tests of every normalization used in the argument.

A A short dependency and status ledger

Statement	Dependency	Status in this article
Original all-orders product expansion	Brent–Glasser–Guttmann, Corollary 9	Published input; cited, not re-proved
Quadratic identity	Elementary polynomial algebra	Fully proved here; symbolic finite identity check included
Integral even r_k	Formal bridge, weight arithmetic, triangular recurrence	Fully proved here for every index
Congruence modulo 32	Evenness, operator valuations, reduction modulo 64	Fully proved here for every index
Polynomial deformation	Parameterized formal equation	Fully proved here; analytic identification only where an expansion is established
Denominator claims	Integrality, mod-32 result, factorial valuations	Fully proved here within the stated binary-digit range
Inverse expansion	Published product expansion and formal reversion	All fixed algebraic orders proved; no effective remainder constants
Negativity for $k \geq 3$	Would require additional sign arguments	Not proved; finite observations only
Exponentially small sectors	Would require further analytic information	Not determined
Lean/Rocq certification	Would require new formal developments and builds	Not performed; a proposed route is described

[Write note, batch 86A, 4 October 2026.] The ledger describes Part I. Negativity for $k \geq 3$ is proved in Part II (Theorem 13.1); exponentially small sectors and Lean/Rocq certification remain as stated.

B Source audit and reproducibility scope

The source audit inspected the final journal PDF of [1], including the conjecture and congruence on its printed page 10; the live A321941 entry; the repository’s transseries README; and the Catalan source module at the pinned commit. Web searches included the exact sequence identifier with “proof”, “integrality”, and “congruence”, as well as the authors and paper title. This is a reproducible targeted audit, not an exhaustive search of all publications, preprints, communications, or repository history.

The package contains the editable L^AT_EX source, the compiled PDF, a standalone verification script, the actual check log, coefficient data, parameter-polynomial data, numerical diagnostics, and source notes. No third-party paper or font file is redistributed. The script writes generated outputs to its own directory by default; use `-out` to select another location. Its main exact arithmetic does not require network access.

[Write note, batch 73O1.] In the collection the compiled PDF is a build of this text, not the delivered one; the script’s own directory is `code/`, so a run without `-out` would write its files there, beside the program, and leave the records in `data/` unchanged. The README gives a rerun on a copy.

Part II

Signs, factorial divergence, and a Borel singularity

13 The sign question and the large-order results

[Write note, batch 86A, 4 October 2026.] This Part answers question 1 of Section 11 (the negativity $r_k < 0$ for every $k \geq 3$) and the growth-law half of question 4 there. It is printed from a later manuscript by the same pipeline; its provenance, status and notation are in Section 13.1. Its first section reproduces the A321941 portions of that manuscript’s opening section, which it shares with the manuscript’s part on OEIS A088714.

The manuscript’s opening reads as follows. It develops two lines of research suggested by the *ProveIt* repository. The first concerns the signs and large-order growth of the asymptotic coefficients underlying OEIS A321941. The second concerns the much finer growth of OEIS A088714, whose defining equation involves composition of its own generating series; that line is printed as Part V of the report `a088714-bell-scale-growth`, not here. The organizing principle is to replace a difficult global coefficient problem by an exact positive representation, and then prove the uniform estimates that the representation requires.

The repository was inspected at the fixed commit

`b7e4f25b6e79cc669fb245cba520017a0f4228b7`.

The two reports used are Part I of the present report and the A088714 report; their paths and Git blob identifiers are recorded in Appendix D. Fixing the snapshot matters: some conjectures in the original publications have already been resolved in these reports, whereas others are explicitly left open.

13.1 Place in the ProveIt collection

[Write note, batch 86A, 4 October 2026.]

Provenance. This Part is built from one manuscript: Part I (Sections 2–7 and Appendix A) of manuscript 02 of batch 86 of ProveIt’s incoming-reports intake, *Signs, Factorial Divergence, and Bell Normalization. New results for OEIS A321941, A088714, and A088713*, dated 4 October 2026, author line “ChatGPT”, in the archive `oeis_research_bundle.zip`. The archive arrived in commit `ae9baa422` and was placed by commit `0f084afa9`; its pin is `b7e4f25b6`, at which this report’s `article.tex` (blob `10873a84`) was the text now printed as Part I. The manuscript’s Part II (its Sections 8–12, the Bell normalization of A088714 and A088713) is printed as Part V of `generating-functions-and-asymptotics/oeis-sequence-asymptotics/a088714-bell-scale-growth`. Its shared material is divided by subject. Here are printed the A321941 portions of its Section 1 (Section 13), its Sections 2–7 (Sections 14–19), the A321941 rows and paragraphs of its Sections 13–14 (Section 20), its research questions 15.6–15.9 (Section 21), its Appendix A (Appendix C) and the A321941 parts of its Appendix B (Appendix D). Nothing was merged: no other manuscript treats A321941. The manuscript’s theorem numbering is replaced by this report’s: its Theorems 1.1, 1.2 and 7.1 are Theorems 13.1, 13.2 and 19.1, and its Sections 2–7 are Sections 14–19.

Edits. The text is printed as delivered, except that every label carries the prefix `bgg:sg:`, the symbols listed in Table 4 as renamed were renamed, references to the manuscript’s other parts, to its source report (now Part I) and to its delivery files were re-pointed, and notes marked [Write

note] were added. A manuscript reference to the BGG article now cites the same bibliography entry as Part I; the manuscript accessed the OEIS entry A321941 on 4 October 2026.

Status. The manuscript is AI-assisted and unrefereed, and nothing in it is formalized in Lean, Rocq or any other proof assistant. Its source audit records internal reviews (“independent analytic normalization, cutoff32, all-orders and Borel reviews passed”), with `external_peer_review`, `proof_assistant_formalized` and `worldwide_priority_exhaustively_verified` all false. Its delivery README states that the infinite sign assertions, asymptotic tails, convergence and error rates are proved in the manuscript and are not inferred from finite tests, that no exhaustive worldwide priority claim is made, and that the deliverable modifies neither the repository nor the OEIS entries. The intake reran the five A321941 programs on a copy; every recorded output was reproduced (up to Windows line endings). Independently of the manuscript’s code, it compared the manuscript’s heat coefficients with Part I’s own coefficient table, computed by a different program from the quadratic master equation: $d_k = (\frac{1}{2})_k h_k$ holds exactly for $0 \leq k \leq 128$. From that table it also recomputed the middle column and the last column of Table 5, confirmed $64^k (\frac{1}{2})_k \gamma_k = S_k$ for $1 \leq k < 60$, $r_k < 0$ for $3 \leq k \leq 256$, and the exact margin 61/105 of (17.6), attained at $k = 3$; and it checked the rational constants of Lemma 16.1 and of Section 17 by hand. It did not re-derive the special-function inputs of Sections 14 and 15.

Shipped layout. The five programs of this Part are shipped as `code/02-sign-*.py` and their recorded outputs as `data/02-sign-*`, with the pinned SymPy requirement as `data/02-sign-requirements-symbolic.txt`. The manuscript’s driver, which also runs the A088714 programs, its two run records and its source audit are shipped once, with Part V of the A088714 report. The paths printed below are the delivered ones; the README gives reruns on copies. The manuscript’s L^AT_EX source and PDF are not shipped and survive in the arrival commit.

Relation to Part I. The sign proof does not use Part I’s integrality theorem. Its comparison scale S_k is exactly $-P_k(0)$ of Theorem 5.2, the constant edge of the polynomial deformation, and Theorem 13.2 shows that this edge is the leading asymptotic scale of $r_k = P_k(1)$: $P_k(1)/P_k(0) \rightarrow 1$. The Borel transform of Section 19 is radial information inside the disk of convergence; continuation and the full singularity set remain open (Section 21), so question 4 of Section 11 is answered in its growth-law part only.

Notation. Part I and this Part share a_n , b_n , ρ_n , d_k , r_k , k , n , and Kummer’s functions M and U , with the same meanings. Every other symbol of this Part has the meaning of Table 4, whose last column gives the tempting false reading.

The regular solution $u_n(r)$ of Section 14 and the scalar $u = 16/k$ of Section 17 share a letter in the manuscript; the subscript and argument distinguish them.

13.2 The A321941 question

Let

$$f_0(z) = \exp\left(\frac{z}{1-z}\right), \quad f_1(z) = \exp\left(\frac{1}{1-z}\right) E_1\left(\frac{1}{1-z}\right),$$

where $E_1(x) = \int_x^\infty e^{-u} u^{-1} du$, and put $a_n = [z^n]f_0(z)$, $b_n = [z^n]f_1(z)$, and $\rho_n = a_n b_n$. Brent, Glasser, and Guttmann [1], abbreviated BGG, proved the Poincare expansion

$$\rho_n \sim -\frac{1}{2n^{3/2}} \sum_{k \geq 0} d_k n^{-k}, \quad d_0 = 1. \quad (13.1)$$

Set $r_k = 64^k d_k$. Strictly speaking, A321941 lists the *numerators* of the rational numbers r_k [2]. The integrality theorem of Part I (Theorem 1.1) identifies these numerators with r_k itself. The sign argument here does not require that integrality theorem: a rational number and its numerator have the same sign.

Table 4: Symbols of Part II that also occur in Part I. Renamed symbols are marked (*ms.* $\cdot$) with the manuscript's letter.

Symbol	Meaning in Part II	<i>Not:</i> meaning in Part I
a_n, b_n	$[z^n]f_0, [z^n]f_1$ (<i>ms.</i> α_n, η_n)	— (same objects)
t	heat-kernel time	the deformation parameter of $P_k(t)$, used here only through $P_k(0), P_k(1)$
$H(t), h_k$	the formal series (15.4) and its coefficients; $d_k = (\frac{1}{2})_k h_k$	the operator $\mathcal{H}$; powers of the formal variable $h = 1/n$ of (3.1)
γ_j	comparison weights (15.3) (<i>ms.</i> β_j)	the inverse coefficients β_k of Theorem 8.1
$q(t)$	$2 \sinh(t/2)/t$	$q_n = 2 + 1/n$
S_k	the scale (13.2), $= -P_k(0)$	the series $S(w), S(z)$ in the proofs of Theorems 5.1–5.2
$\mathcal{M}(t), \mathcal{M}_\ell$	coefficient majorant of Lemma 16.1 (<i>ms.</i> M)	Kummer's $M(a, b, z)$, which keeps its letter
N	fixed order in Theorem 13.2 and Proposition 18.1 (<i>ms.</i> R)	the series $R(z)$ of (3.6)
ε_k	Borel remainder, proof of Theorem 19.1 (<i>ms.</i> e_k)	the weights $e_{j,\ell}$
$G_n(r, s), K(t; 2, 2)$	Green kernel, heat kernel	$G = eE_1(1)$
$F(n), F_m(j)$	$-2n^{3/2}\rho_n; [t^m]\mathcal{A}(t)q(t)^j$	a generic series F ; $F = R^{2/3}$
W_r, W_z	Wronskians of the continuous solutions	the discrete Wronskian W_n
ψ_m, λ_m	eigenfunctions, eigenvalues of $\mathcal{L}$	$\psi(h) = D(h)^{-2/3}$; the parameter λ
u, x, J	$16/k, 1/k$, the split index	a solution $u_n, x_n = na_n$, the polynomial $J(h)$
B, T, E (App. C)	$q^{-1/2}, \tanh(t/4), e^{-T}$	$B(h)$, the array T_n , the operator $E(D)$
$c_j, C_B, \mathcal{B}_2$	large-order coefficients, Borel constant, Borel transform	—

BGG's Remark 11 reports negativity for $3 \leq k \leq 1000$. The repository report (Part I here) proves integrality and congruences, but explicitly leaves negativity at every later index open. We prove:

Theorem 13.1 (Complete sign pattern). *The coefficients in (13.1) satisfy*

$$r_0 = 1, \quad r_1 = -14, \quad r_2 = 86, \quad \boxed{r_k < 0 \text{ for every integer } k \geq 3.}$$

The proof supplies more than eventual negativity. It gives a uniform inequality for every $k \geq 32$, with the remaining 29 cases certified in exact rational arithmetic. An exact spectral identification, proved below, establishes that the coefficients being bounded are precisely those of (13.1).

Define the positive comparison scale

$$S_k = \binom{2k}{k} (2k-3)!! (2k+1)!! \quad (k \geq 1), \quad (13.2)$$

with the convention $(-1)!! = 1$. Our second result describes the expansion *index* k tending to infinity, rather than the original variable n .

Theorem 13.2 (Large-order expansion). *There are effectively computable rational numbers c_j , with $c_0 = 1$, such that, for each fixed $N \geq 0$,*

$$r_k = -S_k \left(\sum_{j=0}^N \frac{c_j}{k^j} + O_N(k^{-N-1}) \right).$$

The first terms are

$$\frac{r_k}{-S_k} = 1 - \frac{1}{3k} - \frac{11}{18k^2} - \frac{1279}{1620k^3} - \frac{5089}{9720k^4} + O(k^{-5}). \quad (13.3)$$

In particular,

$$r_k \sim -\frac{16^k (k!)^2}{\pi^{3/2} k^{3/2}}, \quad d_k \sim -\frac{(k!)^2}{\pi^{3/2} 4^k k^{3/2}}.$$

These estimates determine the exact order of factorial divergence and the first positive singularity of an explicitly normalized Borel transform; see Theorem 19.1.

13.3 What the proofs and computations establish

The mathematical proofs are given in the manuscript. The accompanying programs check exact finite identities, the finite sign certificate, and independently generated asymptotic coefficients. Their role is specific: no finite experiment substitutes for a uniform estimate over an infinite tail. The conclusions are advances beyond the inspected reports and the primary literature cited here. This statement concerns that source boundary; it is not an exhaustive claim of worldwide priority. The manuscript is a research proof with independent checks, rather than a proof-assistant formalization or a claim of external peer review. The repository's existing arithmetic theorems are credited as prior results throughout.

14 An exact radial-oscillator representation

The key step is an exact representation of the product in (13.1). BGG's Lemmas 1–2 give, for $n \geq 1$,

$$a_n = e^{-1} M(n+1, 2, 1), \quad b_n = -\Gamma(n) U(n, 0, 1), \quad (14.1)$$

where M and U are Kummer functions [1, 6].

Consider the Friedrichs realization on $L^2((0, \infty), dr)$ of

$$\mathcal{L} = -\frac{d^2}{dr^2} + \frac{r^2}{16} + \frac{3}{4r^2}. \quad (14.2)$$

The regular solution at zero and recessive solution at infinity of $(\mathcal{L} + n)y = 0$ are, respectively,

$$u_n(r) = r^{3/2} e^{-r^2/8} M(n+1, 2, r^2/4), \quad v_n(r) = r^{3/2} e^{-r^2/8} U(n+1, 2, r^2/4).$$

These statements follow by substituting $z = r^2/4$ in Kummer's equation. The behaviors at the two endpoints also specify the Green kernel directly, if one prefers to start from the differential equation.

The Wronskian identity and transformation

$$W_z\{M(n+1, 2, z), U(n+1, 2, z)\} = -\frac{e^z z^{-2}}{\Gamma(n+1)}, \quad U(n, 0, z) = zU(n+1, 2, z)$$

are DLMF 13.2.34 and 13.2.40 [6]. The first formula uses ordinary M ; the regularized convention in DLMF agrees here because $\Gamma(2) = 1$. Multiplication by the gauge factor squared and $dz/dr = r/2$ gives

$$W_r\{u_n, v_n\} = -\frac{8}{\Gamma(n+1)}.$$

Consequently the kernel of $(\mathcal{L} + n)^{-1}$ is

$$G_n(r, s) = \frac{\Gamma(n+1)}{8} u_n(\min(r, s)) v_n(\max(r, s)).$$

At $r = s = 2$, this becomes

$$G_n(2, 2) = e^{-1} \Gamma(n+1) M(n+1, 2, 1) U(n, 0, 1) = -n \rho_n. \quad (14.3)$$

Thus the exact normalization needed for (13.1) is

$$F(n) := -2n^{3/2} \rho_n = 2\sqrt{n} G_n(2, 2). \quad (14.4)$$

In particular, every constant in the coefficient identification is fixed before any asymptotic expansion is made.

15 The heat kernel and its formal coefficients

15.1 Summing the spectral kernel

The normalized eigenfunctions and eigenvalues of $\mathcal{L}$ are

$$\psi_m(r) = \frac{r^{3/2} e^{-r^2/8}}{\sqrt{8(m+1)}} L_m^{(1)}(r^2/4), \quad \lambda_m = m+1.$$

The Laguerre equation verifies the eigenvalue. Their normalization follows from

$$\int_0^\infty r^3 e^{-r^2/4} L_m^{(1)}(r^2/4)^2 dr = 8 \int_0^\infty x e^{-x} L_m^{(1)}(x)^2 dx = 8(m+1).$$

Laguerre completeness gives a basis [7]. Hence

$$K(t; 2, 2) = e^{-1} \sum_{m \geq 0} \frac{L_m^{(1)}(1)^2}{m+1} e^{-(m+1)t}.$$

For $0 < z < 1$, the Hille–Hardy identity is

$$\sum_{m \geq 0} \frac{L_m^{(1)}(1)^2}{m+1} z^m = \frac{z^{-1/2}}{1-z} \exp\left(-\frac{2z}{1-z}\right) I_1\left(\frac{2\sqrt{z}}{1-z}\right);$$

this is DLMF 18.18.27 with both arguments equal to one [7]. Substitution of $z = e^{-t}$ yields the exact formula

$$K(t; 2, 2) = \frac{\exp[-\coth(t/2)]}{2 \sinh(t/2)} I_1(\operatorname{csch}(t/2)), \quad t > 0. \quad (15.1)$$

The nonnegative spectral summands allow Tonelli’s theorem to give

$$G_n(2, 2) = \int_0^\infty e^{-nt} K(t; 2, 2) dt. \quad (15.2)$$

The heat kernel is $O(t^{-1/2})$ near zero and decays exponentially at infinity, so this integral is finite for $n > 0$.

15.2 A coefficient formula with a negative leading contribution

Introduce

$$q(t) = \frac{2 \sinh(t/2)}{t}, \quad \mathcal{A}(t) = q(t)^{-1/2} e^{-\tanh(t/4)}, \quad \gamma_j = \frac{(2j-3)!!(2j+1)!!}{16^j j!} \quad (j \geq 1). \quad (15.3)$$

The calligraphic symbol $\mathcal{A}$ belongs only to this Part; it is different from the A088714 series A of the manuscript's other part. The large-positive-argument expansion of I_1 is

$$I_1(x) \sim \frac{e^x}{\sqrt{2\pi x}} \left(1 - \sum_{j \geq 1} \frac{(2j-3)!!(2j+1)!!}{8^j j! x^j} \right) \quad [8].$$

Every displayed correction is negative. Using $\coth(t/2) - \operatorname{csch}(t/2) = \tanh(t/4)$, define the formal series

$$H(t) = \mathcal{A}(t) \left(1 - \sum_{j \geq 1} \gamma_j t^j q(t)^j \right) = \sum_{k \geq 0} h_k t^k. \quad (15.4)$$

For each finite order the Bessel expansion in (15.1) proves

$$K(t; 2, 2) \sim (4\pi t)^{-1/2} \sum_{k \geq 0} h_k t^k \quad (t \downarrow 0).$$

Equation (15.4) is a *formal* identity. It does not assert convergence of the infinite Bessel expansion or of H .

Integrating a finite truncation in (15.2) and bounding its remainder gives Watson's lemma [9]. In combination with (14.4), it proves

$$F(n) \sim \sum_{k \geq 0} \frac{\Gamma(k + \frac{1}{2})}{\sqrt{\pi}} h_k n^{-k}.$$

Uniqueness of Poincare coefficients now gives the exact relation

$$\boxed{d_k = \left(\frac{1}{2}\right)_k h_k}. \quad (15.5)$$

In particular, the sign of r_k is the sign of h_k . The first values, directly from (15.4), are

$$h_0 = 1, \quad h_1 = -\frac{7}{16}, \quad h_2 = \frac{43}{1536}, \quad h_3 = -\frac{61}{8192}. \quad (15.6)$$

16 Coefficient majorants

For a formal series P , write $|P|$ for the series of absolute values of its coefficients. We write $P \preceq Q$ when coefficientwise domination holds and Q has nonnegative coefficients.

Lemma 16.1. *The following coefficient inequalities hold:*

$$q(t) \preceq e^{t^2/24}, \quad |\mathcal{A}(t)| \preceq \mathcal{M}(t) := (2 - q(t))^{-1/2} e^{\tan(t/4)}.$$

The series $\mathcal{M}$ has nonnegative coefficients, $\mathcal{M}(0) = 1$, and

$$\mathcal{M}(1) < \frac{3}{2}. \quad (16.1)$$

Moreover $q(1) \leq 24/23$.

Proof. The coefficient of t^{2m} in q is $1/(4^m(2m+1)!)$. Thus the first assertion is equivalent to $6^m m! \leq (2m+1)!$, which follows by induction.

Write $u = q - 1$, so u has nonnegative coefficients. The absolute binomial coefficients of $(1+u)^{-1/2}$ are the coefficients of $(1-u)^{-1/2}$. Similarly, the absolute coefficients of $\tanh(t/4)$ are those of $\tan(t/4)$. Positivity of the latter coefficients also follows recursively from $T' = (1+T^2)/4$, $T(0) = 0$, for $T(t) = \tan(t/4)$. Multiplication and exponentiation preserve coefficient majorants, giving the asserted bound by M .

Here are rational bounds for (16.1). The first nonconstant term in $q(1)$ is $1/24$, and the ratio of successive terms thereafter is at most $1/80$. Therefore

$$q(1) \leq 1 + \frac{1/24}{1 - 1/80} = \frac{247}{237} < \frac{21}{20}.$$

It follows that $(2 - q(1))^{-1/2} < \sqrt{20/19} < 11/10$. The elementary estimates $\sin(1/4) \leq 1/4$ and $\cos(1/4) \geq 31/32$ give $\tan(1/4) \leq 8/31$. Since $e^x \leq (1-x)^{-1}$ for $0 \leq x < 1$,

$$\mathcal{M}(1) < \frac{11}{10} \frac{31}{23} = \frac{341}{230} < \frac{3}{2}.$$

Finally $q(1) \leq e^{1/24} \leq 24/23$. □

Lemma 16.2. *The positive numbers γ_j satisfy*

$$\frac{\gamma_j}{\gamma_{j-1}} = \frac{(2j-3)(2j+1)}{16j} \geq \frac{j}{8} \quad (j \geq 3). \quad (16.2)$$

They decrease through γ_5 and increase thereafter. For every $J \geq 15$, $\gamma_J = \max_{1 \leq j \leq J} \gamma_j$, and

$$\gamma_{16} = \frac{61394155884765868567528125}{604462909807314587353088} > 100.$$

Proof. The ratio follows from (15.3); its lower bound is equivalent to $2j^2 - 4j - 3 \geq 0$ for $j \geq 3$. The same ratio proves the monotonicity assertions. The displayed rational evaluation gives $\gamma_{16} > 100$, while $\gamma_{16}/\gamma_{15} = 957/256 < 4$. Hence $\gamma_{15} > 25 > \gamma_1 = 3/16$, which settles the maximum assertion. □

17 Negativity at every index

17.1 A uniform bound beyond index 31

Extracting the coefficient of t^k in (15.4) gives

$$h_k = -\gamma_k + \mathcal{A}_k - \sum_{j=1}^{k-1} \gamma_j [t^{k-j}] \mathcal{A}(t) q(t)^j, \quad \mathcal{A}_k = [t^k] \mathcal{A}(t). \quad (17.1)$$

We show that the absolute value of all terms other than $-\gamma_k$ is less than $9\gamma_k/10$.

Fix $k \geq 32$, set $u = 16/k \leq 1/2$, and split the sum at $J = \lceil k/2 \rceil - 1$. For the near indices $j \geq \lceil k/2 \rceil$, put $m = k - j$. The ratio bound (16.2) gives $\gamma_{k-m}/\gamma_k \leq u^m$. Also

$$|\mathcal{A}q^j| \leq \mathcal{M}(t) e^{kt^2/24}.$$

Consequently their total normalized absolute contribution is at most

$$\mathcal{M}(u) e^{ku^2/24} - 1. \quad (17.2)$$

Since M has nonnegative coefficients and $u^\ell \leq u$ for $\ell \geq 1$,

$$\mathcal{M}(u) \leq 1 + u(\mathcal{M}(1) - 1) < 1 + \frac{u}{2} = 1 + \frac{8}{k}.$$

Using $ku^2/24 = 32/(3k) < 1$ and $e^x \leq (1 - x)^{-1}$,

$$\mathcal{M}(u)e^{ku^2/24} - 1 < \frac{1 + 8/k}{1 - 32/(3k)} - 1 = \frac{56}{3k - 32} \leq \frac{7}{8}. \quad (17.3)$$

For the far indices $1 \leq j \leq J$, Lemma 16.2 applies because $J \geq 15$. It gives

$$\frac{\gamma_j}{\gamma_k} \leq \frac{\gamma_J}{\gamma_k} \leq (16/k)^{k/2}.$$

Evaluation of the nonnegative majorant at one gives

$$|[t^{k-j}]\mathcal{A}q^j| \leq \mathcal{M}(1)q(1)^j \leq \frac{3}{2}(24/23)^{k/2}.$$

There are fewer than $k/2$ far indices, so their combined normalized contribution is bounded by

$$\frac{3k}{4} \left(\frac{384}{23k} \right)^{k/2} < \frac{1}{100} \quad (k \geq 32). \quad (17.4)$$

For completeness, the expression on the left decreases at successive integers: its ratio is at most $(33/32)\sqrt{12/23} < 1$. At $k = 32$ it equals $24(12/23)^{16} < 1/100$, an exact rational inequality.

Finally

$$\frac{|\mathcal{A}_k|}{\gamma_k} < \frac{3}{200},$$

by (16.1) and $\gamma_k \geq \gamma_{16} > 100$. Combining the three contributions proves

$$\left| \frac{h_k}{\gamma_k} + 1 \right| < \frac{7}{8} + \frac{1}{100} + \frac{3}{200} = \frac{9}{10} \quad (k \geq 32). \quad (17.5)$$

In particular

$$-\frac{19}{10}\gamma_k < h_k < -\frac{1}{10}\gamma_k < 0.$$

17.2 The finite certificate

All h_k in (15.4) are rational. Appendix C gives a short executable certificate for $3 \leq k \leq 31$, using only integer and rational arithmetic. The supplied full verifier checks $3 \leq k \leq 128$; a separately implemented recurrence from BGG agrees with the heat construction through degree 64. The exact minimum in the required finite range is

$$\min_{3 \leq k \leq 31} \frac{-h_k}{\gamma_k} = \frac{61}{105} > 0. \quad (17.6)$$

Thus every finite case is decided by an exact comparison. Together with (17.5), (15.5), and (15.6), this proves Theorem 13.1.

The initial scaled values provide a useful normalization check:

$$\begin{aligned} r_0, \dots, r_6 = & 1, -14, 86, -3660, -1042202, \\ & -247948260, -108448540420. \end{aligned}$$

They agree with the defining OEIS entry [2].

18 The complete expansion as k grows

18.1 A finite formula at each asymptotic order

For fixed $m \geq 0$, define

$$F_m(j) = [t^m] \mathcal{A}(t) q(t)^j.$$

Because $\log q(t)$ starts at degree two, $F_m(j)$ is a polynomial in j of degree at most $\lfloor m/2 \rfloor$. For fixed m and $k > m$,

$$\frac{\gamma_{k-m}}{\gamma_k} = \prod_{r=0}^{m-1} \frac{16(k-r)}{(2(k-r)-3)(2(k-r)+1)}. \quad (18.1)$$

Its order is k^{-m} . Thus the product $(\gamma_{k-m}/\gamma_k)F_m(k-m)$ starts at order $k^{-\lceil m/2 \rceil}$.

Proposition 18.1. *For every fixed integer $N \geq 0$,*

$$\frac{h_k}{-\gamma_k} = \sum_{m=0}^{2N} \frac{\gamma_{k-m}}{\gamma_k} F_m(k-m) + O_N(k^{-N-1}). \quad (18.2)$$

For $m = 0$, the summand is understood to be one.

Proof. Use the same far/near split as in Section 17. The far bound (17.4) is smaller than every power of k^{-1} . The pure analytic contribution also has that property: from (16.2),

$$\gamma_k \geq \frac{\gamma_2}{2} 8^{2-k} k!, \quad |\mathcal{A}_k| < \frac{3}{2}.$$

For the remaining near terms with $m \geq 2N+1$, expand $\mathcal{M}(t) = \sum_{\ell \geq 0} \mathcal{M}_\ell t^\ell$. Their majorant is

$$\sum_{\ell+2v \geq 2N+1} \mathcal{M}_\ell u^\ell \frac{(ku^2/24)^v}{v!}, \quad u = \frac{16}{k}.$$

In terms of $x = 1/k$, this is the nonnegative analytic series

$$\sum_{\ell+2v \geq 2N+1} \mathcal{M}_\ell 16^\ell \left(\frac{32}{3}\right)^v \frac{x^{\ell+v}}{v!}. \quad (18.3)$$

Every exponent satisfies $\ell + v \geq N + 1$. The unconstrained series is $\mathcal{M}(16x)e^{32x/3}$, analytic near zero. Therefore (18.3) is $O_N(x^{N+1})$. Keeping the finitely many remaining terms proves (18.2). $\square$

Expanding the rational functions in (18.1), and the finite polynomials F_m , gives an algorithm for every coefficient in Theorem 13.2. No divergent series is summed in this calculation. Through $m = 8$, it gives

$$\frac{h_k}{-\gamma_k} = 1 - \frac{1}{3k} - \frac{11}{18k^2} - \frac{1279}{1620k^3} - \frac{5089}{9720k^4} + O(k^{-5}).$$

Two independent exact implementations reproduce these coefficients. The primary generator additionally records

$$c_5 = \frac{124787}{81648}, \quad c_6 = \frac{368256067}{36741600}, \quad c_7 = \frac{8769001777}{220449600}, \quad c_8 = \frac{173295857159}{1322697600}.$$

18.2 Factorial scale and numerical diagnostics

The exact identity

$$64^k (\tfrac{1}{2})_k \gamma_k = S_k = \frac{16^k}{\pi^{3/2}} \frac{\Gamma(k - \tfrac{1}{2}) \Gamma(k + \tfrac{1}{2}) \Gamma(k + \tfrac{3}{2})}{\Gamma(k + 1)} \quad (18.4)$$

converts Proposition 18.1 into Theorem 13.2. Stirling's formula gives

$$S_k \sim \frac{16^k (k!)^2}{\pi^{3/2} k^{3/2}}, \quad \frac{r_{k+1}}{r_k} \sim 16k^2.$$

Thus both ordinary formal series $\sum d_k z^k$ and $\sum r_k z^k$ have radius zero and exact Gevrey order two: their coefficients grow at the scale of a fixed exponential times $(k!)^2$, and no lower factorial order can bound them.

Table 5 illustrates the proved expansion. Each entry in its middle column is rounded only after exact rational evaluation. The last column subtracts the displayed four-correction approximation (13.3). These values check the scale and normalization; the uniform remainder was proved above.

Table 5: Large-order diagnostics for A321941.

k	$r_k/(-S_k)$	Difference from (13.3)
3	0.580952380952	$-2.0433 \cdot 10^{-1}$
4	1.050342151675	$1.8625 \cdot 10^{-1}$
8	0.947282773338	$1.6788 \cdot 10^{-4}$
16	0.976581013320	$2.2388 \cdot 10^{-6}$
32	0.988962008187	$5.6170 \cdot 10^{-8}$
64	0.994639428263	$1.5788 \cdot 10^{-9}$
128	0.997358155702	$4.6832 \cdot 10^{-11}$

19 An order-two Borel singularity

We fix the factorial convention explicitly:

$$\mathcal{B}_2(s) = \sum_{k \geq 0} \frac{d_k}{(2k)!} s^k. \quad (19.1)$$

Theorem 19.1. *The power series (19.1) has radius of convergence exactly 16. There is a finite real constant C_B such that*

$$\mathcal{B}_2(16x) = \frac{1}{\pi} \log(1-x) + C_B + O((1-x)|\log(1-x)|) \quad (x \uparrow 1). \quad (19.2)$$

In particular the positive boundary point $s = 16$ is singular.

Proof. The gamma duplication formula and (18.4) give

$$\frac{(\tfrac{1}{2})_k \gamma_k}{(2k)!} = \frac{\Gamma(k - \tfrac{1}{2}) \Gamma(k + \tfrac{3}{2})}{\pi 16^k \Gamma(k + 1)^2}.$$

Theorem 13.2 therefore yields

$$\frac{16^k d_k}{(2k)!} = -\frac{1}{\pi k} + O(k^{-2}).$$

The root test proves the asserted radius. Let

$$\varepsilon_k = \frac{16^k d_k}{(2k)!} + \frac{1}{\pi k}, \quad C_B = 1 + \sum_{k \geq 1} \varepsilon_k.$$

The constant is well defined because $\sum |\varepsilon_k| < \infty$. For $0 < x < 1$, use $-\sum_{k \geq 1} x^k/k = \log(1-x)$. The remaining boundary error is bounded by a constant times

$$\sum_{k \geq 1} \frac{1-x^k}{k^2} = O((1-x)|\log(1-x)|).$$

To verify the last estimate directly, split at $k = \lfloor (1-x)^{-1} \rfloor$ and use $1-x^k \leq \min\{k(1-x), 1\}$. $\square$

This is a radial boundary statement proved inside the convergence disk. It makes no claim of analytic continuation or Borel summability. With the same divisor $(2k)!$, the transform of $r_k = 64^k d_k$ has radius $1/4$, rather than 16 . The change of radius is entirely due to the normalization.

20 What has changed, and what the programs check

[Write note, batch 86A.] This section prints the A321941 portions of the manuscript's Sections 13 and 14, which are shared with its A088714 part. The three A088714 and A088713 rows of its table, its paragraph on A088714, its A088714 check items and its normalization diagnostics are printed in Part V of the A088714 report.

20.1 What has changed mathematically

The source boundary and the new conclusions for A321941 are summarized in Table 6. The strongest improvements concern infinite signs and multiplicative normalization. In the first problem, integrality alone gives little control over sign. In the second, an n -th-root asymptotic leaves room for multiplicative factors whose logarithms diverge. The new proofs supply exactly the missing global control in each case.

Table 6: Earlier results and the conclusions established here (A321941 rows of the manuscript's table).

Topic	Inspected earlier boundary	This article
A321941 signs	BGG observed negativity through 1000; the repository leaves the infinite assertion open.	Every $r_k < 0$ for $k \geq 3$, with an explicit bound for all $k \geq 32$.
A321941 growth	The repository gives an exact constant-edge expression for its polynomial deformation.	That expression is the leading asymptotic scale, with a complete inverse- k expansion and an order-two Borel boundary law.

For A321941 the spectral representation turns the sign question into domination by a single explicit coefficient. Near offsets determine the entire large-order expansion, while far offsets are negligible beyond every algebraic order. The formal-series viewpoint is essential: the divergent series is used coefficient by coefficient, after the exact resolvent and heat kernel have already been established.

20.2 What is checked exactly

The primary verification programs use the Python standard library. A separate optional symbolic cross-check uses SymPy. The A321941 routines use `fractions.Fraction`; their sign decisions never involve floating-point arithmetic. The included A321941 checks cover:

1. The required finite sign range $3 \leq k \leq 31$, with an extended check through 128.
2. Agreement with BGG’s published coefficient recurrence through degree 64, and agreement with all 13 initial terms displayed in the inspected OEIS entry.
3. The rational constants in the cutoff-32 proof, including $24(12/23)^{16} < 1/100$ and the combined error $9/10 < 1$.
4. Independent constructions of the large-order coefficients through c_8 .

The tests are finite certificates and checks of algebraic identifications. They do not establish the infinite asymptotic remainder bounds; those were proved in Sections 14–19 (and, for A088714, in Part V of that report). The independent A321941 implementation starts from BGG’s polynomial coefficient recurrence and reconstructs the heat coefficients by a different route. This checks more than agreement of two instances of the same series-expansion code.

20.3 Rebuilding the deliverable

The archive contains a self-contained `oeis_research.tex`, its compiled PDF, the verification code, exact coefficient data, recorded outputs, and a source audit. The TeX source has no external figures or bibliography file. Compile it with a standard TeX Live installation:

```
latexmk -pdf -interaction=nonstopmode -halt-on-error oeis_research.tex
```

The file `README.txt` gives the individual verification commands. Python 3.11 or later supports the Decimal methods used by the optional normalization diagnostics. The exact rational sign checks and integer endpoint checks need no third-party numerical package.

[Write note, batch 86A.] The delivered `oeis_research.tex`, its PDF and its `README.txt` are not shipped; they survive in the arrival commit `ae9baa422`. In the collection the A321941 programs are `code/02-sign-finite_certificate.py`, `02-sign-verify_sign.py`, `02-sign-independent_check.py`, `02-sign-large_order.py` and `02-sign-asymptotic_check.py`, and their recorded outputs are in `data/` under the same prefix. Two of them find a sibling by its delivered name (`large_order.py` imports `verify_sign`, `asymptotic_check.py` loads `independent_check.py`), four write their output beside themselves, and the manuscript’s driver `run_checks.py`, shipped with Part V of the A088714 report, expects the delivered layout and rewrites its own run records. The README of this report therefore gives reruns on copies with the delivered names. The PDF of this report is a build of this text.

21 Further research questions for Part II

[Write note, batch 86A.] These are the manuscript’s research questions 15.6–15.9; its questions 15.1–15.5 concern A088714 and are printed in Part V of the A088714 report. Question 21.1 continues question 6 of Section 11, question 21.2 is the part of question 4 there that this Part leaves open, and question 21.3 bears on questions 2 and 3 there.

The following questions are deliberately more precise than a request for additional numerical terms. Each identifies a mathematical target that is not established by the present proofs.

Research question 21.1 (The entire parameter-dependent sign region). For the radial oscillator evaluated at a general positive point, or for the polynomial deformation in Part I, which parameters give a complete sign pattern analogous to A321941?

The method of Sections 15–17 suggests uniform coefficient majorants on compact parameter sets. The interesting targets are an explicit parameter-dependent cutoff, the location of the exceptional small indices, and a sharp boundary between sign regimes. A separate algebraic problem is to relate that boundary to the integer polynomial family already constructed in the repository.

Research question 21.2 (Complex continuation and exponentially small terms). What is the analytic continuation of $\mathcal{B}_2(s)$ beyond its disk of convergence, and how do its singularities determine the exponentially small terms associated with (13.1)?

Theorem 19.1 proves a logarithmic singularity at $s = 16$ from inside the disk. It does not determine the full singularity set, lateral continuations, or Stokes constants. The exact heat integral is a natural starting point for such an analysis. Care is needed with the order-two factorial convention and with the distinction between large-order coefficients and a remainder theorem for an optimally truncated n -expansion.

Research question 21.3 (Arithmetic beyond the known congruence). Can the complete sign and large-order theorems be combined with the repository’s arithmetic recurrences to determine further exact valuations and congruences of r_k ?

The earlier report already proves integrality, the modulo-32 congruence, and sharp denominator conclusions in specified cases (Part I). Those are not new claims here. The analytic scale can organize searches for additional arithmetic patterns, but real asymptotics cannot by themselves determine a 2-adic valuation. A proof would need a strengthened exact recurrence or an appropriate congruence invariant.

Research question 21.4 (A formal proof with a small trusted certificate). Which portions can be formalized first while keeping the analytic dependency chain explicit?

The finite rational certificate, coefficientwise majorants, and positive-kernel summation identity are good initial targets. The exact spectral bridge requires a larger special-function and operator-analysis library. A useful staged formalization would state the analytic inputs as explicit theorems, verify every finite truncation algebraically, and then discharge the limiting estimates. This would also make the distinction between formal series identities and convergent analytic identities mechanically visible.

[Write note, batch 86A.] The positive-kernel summation identity named in the last question belongs to the manuscript’s A088714 proof (Part V of that report); for A321941 the targets are the certificate of Appendix C and the majorants of Section 16.

C A short exact certificate for the finite sign range

The following complete Python program proves the finite comparisons used in Section 17. It constructs $B = q^{-1/2}$ from $2qB' + q'B = 0$, constructs $T = \tanh(t/4)$ from $T' = (1 - T^2)/4$, and constructs $E = e^{-T}$ from $E' = -T'E$. Truncated multiplication then implements (15.4). Every operation is exact.

```
from fractions import Fraction as F
from math import factorial
N = 31
```

```

def mul(a, b):
    c = [F(0)] * (N + 1)
    for i, x in enumerate(a):
        for j, y in enumerate(b[:N + 1 - i]):
            c[i + j] += x * y
    return c

q = [F(0)] * (N + 1)
for j in range(N // 2 + 1):
    q[2*j] = F(1, 4**j * factorial(2*j + 1))
B = [F(1)]
for k in range(1, N + 1):
    B.append(-sum((F(k)-F(i, 2))*q[i]*B[k-i]
                  for i in range(1, k+1)) / k)
T = [F(0), F(1, 4)] + [F(0)] * (N - 1)
for k in range(2, N + 1):
    T[k] = -sum(T[i]*T[k-1-i] for i in range(k)) / (4*k)
E = [F(1)]
for k in range(1, N + 1):
    E.append(-sum(i*T[i]*E[k-i] for i in range(1, k+1))/k)
A = mul(B, E)
v = [F(0)] + q[:N]
power = [F(1)] + [F(0)] * N
S = power.copy()
b = F(3, 16)
for j in range(1, N + 1):
    if j > 1:
        b *= F((2*j-3)*(2*j+1), 16*j)
    power = mul(power, v)
    for k in range(j, N + 1):
        S[k] -= b * power[k]
h = mul(A, S)
assert h[:4] == [1, F(-7, 16), F(43, 1536), F(-61, 8192)]
assert all(h[k] < 0 for k in range(3, N + 1))
print("Exact certificate: h_k < 0 for 3 <= k <= 31.")

```

[Write note, batch 86A.] This program is shipped as `code/02-sign-finite_certificate.py`, byte for byte the listing above. It writes nothing; it prints one line.

D Source and dependency audit (Part II)

The research reports were retrieved from the public repository at commit

b7e4f25b6e79cc669fb245cba520017a0f4228b7,

inspected on 4 October 2026. The main published source is the final 2019 BGG article [1]; numbering in this manuscript refers to that final version.

A321941 source

The report *Proving Two Arithmetic Conjectures for OEIS A321941* is at

SetTheory/Cardinals/docs/reports/congruences-and-valuations/a321941-asymptotic-coefficient-integrality/article.tex.

Its Git blob identifier is

10873a84fcf7a5f725ad74f90e2bd71ab7e7dd32.

Its proved integrality and congruence results are prior inputs to the source comparison, but are not needed by the sign proof. Its polynomial constant term equals $-S_k$. The contribution here is

to prove the complete sign assertion, identify that expression as the large-order scale, and derive the all-orders and Borel conclusions.

[Write note, batch 86A.] That report is Part I of the present one; the blob is its `article.tex` as it stood before this Part was added. The manuscript’s second source subsection, on the A088714 report, is printed in Part V of that report.

Verification boundaries

The analytic and combinatorial arguments received separate independent checks, including normalization constants, index partitions, all-orders truncations, and the two infinite sums in the convergence argument. The executable checks and their recorded outputs are included in the archive. This internal review does not constitute external refereeing or a proof-assistant certificate. No OEIS entry or repository commit is changed by this deliverable.

[Write note, batch 86A.] “Internal” and “independent” refer to reviews within the manuscript’s own pipeline, recorded in its source audit (shipped as `data/05-bell-norm-SOURCE_AUDIT.json` of the A088714 report). The two infinite sums of the convergence argument belong to its A088714 part.

References

- [1] R. P. Brent, M. L. Glasser, and A. J. Guttmann, *A conjectured integer sequence arising from the exponential integral*, Journal of Integer Sequences **22** (2019), Article 19.4.7. [Journal page](#); [final journal PDF](#); [arXiv:1812.00316](#). Theorem numbering in this article follows the journal version.
- [2] The OEIS Foundation Inc., *A321941: Scaled numerators in the asymptotic expansion of the Maclaurin coefficients in a Hadamard product involving the exponential integral*. Entry contributed by Richard P. Brent, 2018. <https://oeis.org/A321941>. Inspected 1 October 2026.
- [3] The OEIS Foundation Inc., *A000262*, partitions into linearly ordered blocks (“sets of lists”). <https://oeis.org/A000262>. Inspected 1 October 2026.
- [4] V. Reshetnikov and contributors, *ProveIt*, `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/README.md`. Pinned commit: `0a6d798c5f39775b2fb4d85fef643fef996c87cf`. [Pinned repository source](#). Inspected 1 October 2026. The README, rather than the complete consolidated volume, was used for the targeted comparison.
- [5] V. Reshetnikov and contributors, *ProveIt*, `Analysis/FabiusFunction/Lean/FabiusFunction/QuadraticCoreCatalan.lean`, same commit as [4]. [Pinned source module](#). Source inspected; no new formal build performed for this article.
- [6] NIST Digital Library of Mathematical Functions, *Section 13.2: Definitions and Basic Properties*, especially 13.2.34 and 13.2.40. <https://dlmf.nist.gov/13.2>. Accessed 4 October 2026.
- [7] NIST Digital Library of Mathematical Functions, *Sections 18.3 and 18.18*, Laguerre orthogonality, completeness, and the Hille–Hardy formula 18.18.27. <https://dlmf.nist.gov/18.3>; <https://dlmf.nist.gov/18.18.E27>. Accessed 4 October 2026.

- [8] NIST Digital Library of Mathematical Functions, *Sections 10.17 and 10.40*, asymptotic coefficients and large-argument expansions of modified Bessel functions. <https://dlmf.nist.gov/10.17.E1>; <https://dlmf.nist.gov/10.40.E1>. Accessed 4 October 2026.
- [9] NIST Digital Library of Mathematical Functions, *Section 2.3(ii): Watson's Lemma*. <https://dlmf.nist.gov/2.3.ii>. Accessed 4 October 2026.